

a Perturbation enters as data

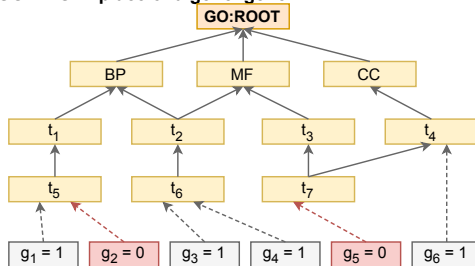
strain = wildtype with $p = \{g_2, g_5\}$ deleted
 gene state $s_i = 0$ if $g_i \in p$, else 1

0 deleted 1 present

The reference's (term, gene) annotation table (59,986 rows) is copied per strain and the rows of deleted genes are zeroed. The strain is a rebuilt data object; no operator acts on a shared encoding H .

Each subsystem reads the states of its own annotated genes and the outputs of its child subsystems; every gene is in at least two subsystems.

GO DAG in place of a gene—gene $A^{(k)}$



edges: is_child_of (child $\rightarrow$ parent); dashed: gene states $s_{\text{genes}(t)}$ entering the subsystems that annotate them; subsystems run by stratum, leaves first (12 $\rightarrow$ 0)

Readout and losses

subsystem t (one per GO term)

$$l_t = [O_c \text{ for children } c \text{ of } t \parallel s_{\text{genes}(t)}]$$

$$O_t = \tanh(\text{BN}(W_t l_t + b_t)) \in [-1, 1]^{L_t}$$

$$L_t = \max(20, \lceil 0.3 \cdot |\text{genes}(t)| \rceil)$$

$$\text{root readout } \hat{y} = w_r^T O_{\text{ROOT}} + b_r \text{ (trigenic interaction } y_{\text{int}})$$

$$\text{auxiliary heads on every } t \neq r: \hat{y}_t = w_t^T O_t + b_t$$

$$\text{loss} = \text{MSE}(\hat{y}, y) + \alpha \cdot \text{mean}_{t \neq r} \text{MSE}(\hat{y}_t, y), \alpha = 0.3$$

$$\text{weight decay in AdamW replaces } \lambda \|W\|_2; \text{init } U(-0.001, 0.001)$$

built on the filtered DAG: 2,655 subsystems, 3,208 edges, 13 strata, 20.6 M parameters

